

$\det \mathcal{G}'_h = 1$. Thus the asymmetric variant of the Euler method is area preserving, even though the standard Euler method is not.

It would be valuable to have a general method for deriving volume conserving methods. It turns out that volume conservation is, itself, most readily obtained as a consequence of a more fundamental property of Hamiltonian flows, the conservation of the symplectic form.

2.3.3 The Symplectic Property

Let $m = 2N_c$. A symplectic map $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is one that preserves the symplectic differential 2-form. The simplest way to write this is as the following algebraic condition on the Jacobian matrix of Φ :

$$\Phi'^T J \Phi' = J.$$

Recall that the Jacobian matrix is the $m \times m$ matrix of partial derivatives of the components of Φ , $(\Phi')_{ij} = \frac{\partial \Phi_i}{\partial z_j}$, while

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}.$$

So the equation $\Phi'^T J \Phi' = J$ represents a large number ($4N_c^2$) of equations.

To explain the origin of this formula, we need a brief diversion into the world of differential forms. This language will also be useful in Chap. 4, where we consider constrained systems.

A 1-form α defined on $\mathbb{R}^m$ is a family of linear mappings from $\mathbb{R}^m$ to $\mathbb{R}$, defined for each point of $\mathbb{R}^m$. Let $\mathbf{a} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, then we may define a one-form associated to this vector by $\alpha(\mathbf{x})(\xi) = \mathbf{a}(\mathbf{x})^T \xi$.

The differential of a function $g : \mathbb{R}^m \rightarrow \mathbb{R}$, denoted dg , is a family of linear mappings (one for each point in phase space) from vectors $\xi \in \mathbb{R}^m$ into the reals defined by

$$dg(\mathbf{q}, \mathbf{p})(\xi) = \nabla g(\mathbf{q}, \mathbf{p})^T \xi.$$

So, denoting the i th position coordinate by q_i , we have $dq_i(\xi) = \xi_i$; the differential is thus an example of a 1-form.

On the other hand the so called *wedge product* of 1-forms α , β , is an example of a 2-form, at any point in phase space it can be viewed as a quadratic form, i.e. a scalar valued function of two vectors which is linear in each argument. It is written $\alpha \wedge \beta$ and is defined, for vectors $\xi, \eta \in \mathbb{R}^m$ by

$$\alpha \wedge \beta(\xi, \eta) = \alpha(\xi)\beta(\eta) - \alpha(\eta)\beta(\xi).$$

The wedge product of the coordinate differentials dq_i, dp_i may be written

$$dq_i \wedge dp_i(\xi, \eta) = \xi_i \eta_{i+N_c} - \xi_{i+N_c} \eta_i = \xi^T J^{(i)} \eta,$$

where $J^{(i)}$ is the matrix which has zeros everywhere except for $(J^{(i)})_{i,i+N_c} = 1$, $(J^{(i)})_{i+N_c,i} = -1$. Summing these terms results in the *symplectic 2-form*, denoted ψ_S :

$$\psi_S = \sum_{i=1}^{N_c} dq_i \wedge dp_i(\xi, \eta) = \xi^T \left(\sum_{i=1}^{N_c} J^{(i)} \right) \eta = \xi^T J \eta.$$

Thus the matrix

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}$$

defines a representation for the 2-form ψ_S . It is easy to verify that

$$\psi_S = \sum_{i < j} dz_i \wedge J_{ij} dz_j.$$

In general, a differential 2-form ψ is represented in coordinates by

$$\psi_z = \sum_{i,j} a_{ij}(z) dz_i \wedge dz_j,$$

with matrix of coefficients $A(z) = (a_{ij}(z))$.

The *pull-back* of a differential 1-form ψ under a phase space mapping Φ is defined as the action of ψ after transformation by the Jacobian matrix of the mapping. It is written $\Phi^* \psi$, so

$$(\Phi^* \psi_z)(\xi) = \psi_{\Phi(z)}(\Phi'(z)\xi).$$

The pull-back of a differential 2-form $\psi_1 \wedge \psi_2$ is consequently defined as

$$\Phi^*(\psi_1 \wedge \psi_2) = (\Phi^* \psi_1) \wedge (\Phi^* \psi_2).$$

Given a differential 2-form ψ_z represented by the matrix $A(z) = (a_{ij}(z))$, the *pull-back* of ψ_z under Φ is defined by

$$\Phi^* \psi_z = \sum_{i,j} b_{ij}(z) dz_i \wedge dz_j,$$